

CentreCOM® GS980MX Series

Stackable Multi-Gigabit Layer 3 Lite Switches



The Allied Telesis Centre COM GS980MX Series of Layer 3 Gigabit switches feature high-capacity, resiliency and easy management. Power over Ethernet models with Multi-Gigabit support make them an ideal solution for high-speed connectivity at the network edge.

Overview

The GS980MX Series are high-performing, feature-rich, and versatile for today's networks. With Gigabit/Multi-Gigabit ports and 10 Gigabit uplinks, plus the power of Allied Telesis Virtual Chassis Stacking (VStack™), the GS980MX Series enable flexible deployment and a resilient solution.

The Power over Ethernet models offer 2.5G and 5G Multi-Gigabit ports to support connecting and powering high-speed Wi-Fi 6 wireless networks, and other high bandwidth applications. The HSm models can provide up to 90 Watts (PoE++) per port. This enables powering high power devices such as high resolution PTZ cameras with heater/blowers for outdoor applications, enhanced infrared lighting, and more.

SPECIFICATIONS

Performance

- 10KB L2 and 9KB L3 Jumbo frames
- 4094 configurable VLANs
- Up to 16K MAC addresses
- 1GB DDR3 SDRAM
- Packet buffer memory:
 - 1.5MB (28-port models)
 - 3MB (10, 18, 52-port models)
 - 4.5MB (52PSm models)

Reliability

- Modular AlliedWare Plus operating system
- Full environmental monitoring of PSUs, fans, temperature and internal voltages.
- MTBF (Mean Time Between Failures)
 - GS980MX/10HSm: 130,000 hours
 - GS980MX18HSm: 110,000 hours
 - GS980MX/28: 290,000 hours
 - GS980MX/28PSm: 200,000 hours
 - GS980MX/52: 210,000 hours
 - GS980MX/52PSm: 150,000 hours

Expandability

- Stack up to 4 units in a VStack at any port speed

Diagnostic Tools

- Active Fiber Monitoring detects tampering on optical links
- Built-In Self test (BIST)
- Cable fault locator (TDR)
- Find-me device locator
- Automatic link flap detection and port shutdown
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling for IPv4 and IPv6
- Port mirroring
- Trace Route for IPv4 and IPv6
- Uni-Directional Link Detection (UDLD)

IPv6 Features

- Equal Cost Multi Path (ECMP) routing
- Static routing and RIP for IPv4
- Static routing for IPv6
- Device management over IPv6 networks with SNMP, Telnet, SSH
- IPv6 hardware ACLs
- Log to IPv6 hosts with Syslog
- IPv6 Ready certified

Management

- Front panel 7-segment LED provides at-a-glance status and fault information
- Autonomous Management Framework Plus (AMF Plus) enables powerful centralized management and zero-touch device installation and recovery
- Manage the GS980MX with Vista Manager EX, our graphical single-pane-of-glass monitoring and management tool for AMF Plus networks. Also supports wireless and third party devices
- From AW+ 5.5.2-2, an AMF Plus license operating in the network provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)
- AMF Security (AMF-Sec) enables a self-defending network—managing the GS980MX Series (or other AMF Plus switches) to automatically block the spread of malware by quarantining suspect end user devices
- IPFIX exports IP flow data in a network for analysis, so administrators have information for accounting, capacity planning, and optimization
- Console management port on the front panel for ease of access
- NETCONF/RESTCONF northbound interface with YANG data modelling
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- Industry-standard CLI with context-sensitive help
- Powerful CLI scripting engine with built-in text editor
- Web-based Graphical User Interface (GUI)

Key Features

- AlliedWare Plus Enterprise-class operating system
- Autonomous Management Framework™ Plus (AMF Plus) edge node
- Vista Manager EX compatible
- AMF-Security compatible
- VStack™ up to 4 switches
- VStack LD for long distance stacking
- EPSR transit node
- 10 Gigabit uplinks
- 2.5/5G with PoE for high-speed wireless APs (PSm and HSm models)
- IEEE 802.3at PoE+ (30W per port on PSm models)
- IEEE 802.3bt PoE++ (90W per port on HSm models)
- Continuous PoE
- Active Fiber Monitoring (AFM)
- IPv6 features
- Eco-Friendly
- Device GUI for web-based management
- NETCONF/RESTCONF with YANG data modelling
- IPFIX (IP Flow Information Export)
- Enhanced Transmission Selection (ETS)
- USB interface allows software release files, configurations and other files to be stored for backup and distribution to other devices
- Comprehensive SNMP MIB support for standards based device management
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- Wirespeed forwarding

Product Specifications

	10/100/1000T (RJ-45) copper ports	100M/1/2.5/5 Gigabit ports	1/10 Gigabit SFP+ ports	Stacking ports	PoE enabled ports	Switching Fabric	Forwarding Rate
GS980MX/10HSm	-	8	2	2*	8	120Gbps	89.2Mpps
GS980MX/18HSm	-	16	2	2*	16	200Gbps	148.8Mpps
GS980MX/28	24	-	4	2*	-	128Gbps	95.2Mpps
GS980MX/28PSm	20	4	4	2*	24	160Gbps	119Mpps
GS980MX/52	48	-	4	2*	-	176Gbps	130.9Mpps
GS980MX/52PSm	40	8	4	2*	48	240Gbps	179Mpps

* Any port/s can be used for stacking

Physical Specifications

	Width	Depth	Height	Mounting	Unpackaged Weight	Packaged Weight	Packaged dimensions
GS980MX/10HSm	210 mm (8.26 in)	362 mm (14.25 in)	42.5 mm (1.67 in)	Rack-mount	3.5 kg (7.7 lb)	5.5 kg (12.1 lb)	461 x 371 x 153 mm (18.15 x 14.60 x 6.02 in)
GS980MX/18HSm	441 mm (17.36 in)	256 mm (10.07 in)	44 mm (1.73 in)	Rack-mount	4.3 kg (9.48 lb)	5.8 kg (12.78 lb)	526 x 412 x 129 mm (20.70 x 16.22 x 5.07 in)
GS980MX/28	441 mm (17.36 in)	323 mm (12.72 in)	44 mm (1.73 in)	Rack-mount	4.3 kg (9.5 lb))	6.3 kg (13.8 lb)	563 x 534 x 128 mm (22.16 x 21.02 x 5.04 in)
GS980MX/28PSm	441 mm (17.36 in)	421 mm (16.57 in)	44 mm (1.73 in)	Rack-mount	5.6 kg (12.4 lb)	7.6 kg (16.7 lb)	563 x 534 x 128 mm (22.16 x 21.02 x 5.04 in)
GS980MX/52	441 mm (17.36 in)	323 mm (12.72 in)	44 mm (1.73 in)	Rack-mount	4.8 kg (10.1 lb)	6.8 kg (14.9 lb)	563 x 534 x 128 mm (22.16 x 21.02 x 5.04 in)
GS980MX/52PSm	441 mm (17.36 in)	421 mm (16.57 in)	44 mm (1.73 in)	Rack-mount	6.1 kg (17.8 lb)	8.1 kg (17.8 lb)	563 x 632 x 128 mm (22.16 x 24.88 x 5.04 in)

Power and Noise Characteristics

	No PoE Load			Full PoE+ Load			Max PoE Power	PoE sourcing ports				
	Max Power Consumption	Max Heat Dissipation	Noise	Max Power Consumption	Max Heat Dissipation	Noise (dBA)		PoE (7.54W)	PoE (15.4W)	PoE+ (30W)	PoE++ (60W)	PoE++ (90W)
GS980MX/10HSm	60	204	42	605	2065	64	500W	8	8	8	8	5
GS980MX/18HSm	86	293	42	970	3317	42	720W	16	16	16	12	8
GS980MX/28	39	133	42*	-	-	-	-	-	-	-	-	-
GS980MX/28PSm	70	239	42*	510	1741	42*	370W	24	24	12	-	-
GS980MX/52	60	205	42*	-	-	-	-	-	-	-	-	-
GS980MX/52PSm	95	324	42*	530	1809	42*	370W	48	24	12	-	-

* This figure is under 30 degree C ambient temperature

Noise: tested to ISO7779; front bystander position

Latency (microseconds)

	100Mbps	1Gbps	2.5Gbps	5Gbps	10Gbps
GS980MX/10HSm	8.24µs	7.89µs	5.63µs	3.49µs	2.12µs
GS980MX/18HSm	8.35µs	7.96µs	7.72µs	5.23µs	2.56µs
GS980MX/28	8.29µs	7.63µs	-	-	1.63µs
GS980MX/28PSm	8.29µs	7.63µs	7.41µs	4.97µs	1.63µs
GS980MX/52	8.34µs	7.75µs	-	-	1.67µs
GS980MX/52PSm	8.34µs	7.75µs	7.51µs	5.06µs	1.67µs

Quality of Service

- 8 priority queues with a hierarchy of high priority queues for real time traffic, and mixed scheduling, for each switch port
- Limit bandwidth per port or per traffic class down to 64kbps
- Wirespeed traffic classification with low latency essential for VoIP and real-time streaming media applications
- IPv6 QoS support
- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Extensive remarking capabilities
- Taildrop for queue congestion control
- Queue scheduling options for Strict priority, weighted round robin or mixed scheduling

- Type of Services (ToS) IP precedence and DiffServ marking based on layer 2, 3 and 4 headers
- Enhanced Transmission Selection (ETS)

Resiliency Features

- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- Dynamic link failover (host attach)
- EPSRing™ (Ethernet Protection Switched Rings) with Super-Loop Protection (SLP) and enhanced recovery for extra resiliency
- Long-Distance VSTACK with fiber modules (VSTACK LD)
- Loop protection: loop detection and thrash limiting
- PVST+ compatibility mode
- STP root guard
- VSTACK fast failover minimizes network disruption

Security Features

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Configurable auth-fail and guest VLANs
- Authentication, Authorization and Accounting (AAA)
- Bootloader can be password protected for device security
- BPDU protection
- DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)
- DoS attack blocking and virus throttling
- Dynamic VLAN assignment
- MAC address filtering and MAC address lock-down
- Network Access and Control (NAC) features manage endpoint security
- Port-based learn limits (intrusion detection)
- Private VLANs provide security and port isolation for multiple customers using the same VLAN
- Secure Copy (SCP)
- Secure File Transfer (SFTP) client
- Strong password security and encryption
- Simple Certificate Enrollment Protocol (SCEP) supports secure management
- Tri-authentication: MAC-based, web-based and IEEE 802.1x
- Web-based authentication

Environmental specifications

- Operating temperature range:
0°C to 50°C (32°F to 122°F)
Derated by 1°C per 305 meters (1,000 ft)
- Storage temperature range:
-25°C to 70°C (-13°F to 158°F)
- Operating relative humidity range:
5% to 90% non-condensing
- Storage relative humidity range:
5% to 95% non-condensing
- Operating altitude range:
Up to 3,000 meters maximum (9,843 ft)

Electrical approvals and compliances

- EMC: EN55024 FCC Class A, EN55032 Class A, EN61000-3-2, EN61000-3-3, VCCI Class A, RCM
- Immunity: EN55024, EN61000-3-levels 2 (Harmonics), and 3 (Flicker) – AC models only

Safety

- Standards: UL60950-1, CSA-C22.2 No. 60950-1, EN60950-1, UL62368-1
- Certifications: cUL, cULus, TUV

Restrictions on Hazardous Substances (RoHS) Compliance

- EU RoHS compliant
- China RoHS compliant

STANDARDS & PROTOCOLS

Authentication

- RFC 1321 MD5 Message-Digest algorithm
- RFC 1828 IP authentication using keyed MD5

Cryptographic Algorithms

FIPS Approved Algorithms

Encryption (Block Ciphers):

- AES (ECB, CBC, CFB and OFB Modes)
- 3DES (ECB, CBC, CFB and OFB Modes)

Block Cipher Modes:

- CCM
- CMAC
- GCM
- XTS

Digital Signatures & Asymmetric Key Generation:

- DSA
- ECDSA
- RSA
- Secure Hashing:
SHA-1
SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512)
Message Authentication:
HMAC (SHA-1, SHA-2(224, 256, 384, 512)
Random Number Generation:
DRBG (Hash, HMAC and Counter)

Non FIPS Approved Algorithms

- RNG (AES128/192/256)
- DES
- MD5

Ethernet Standards

- IEEE 802.2 Logical Link Control (LLC)
- IEEE 802.3 Ethernet
- IEEE 802.3ab1000BASE-T
- IEEE 802.3ae10 Gigabit Ethernet
- IEEE 802.3af Power over Ethernet (PoE)
- IEEE 802.3at Power over Ethernet up to 30W (PoE+)
- IEEE 802.3azEnergy Efficient Ethernet (EEE)
- IEEE 802.3bt Power over Ethernet up to 90W (PoE++)
- IEEE 802.3bz2.5GBASE-T and 5GBASE-T ("multi-gigabit")
- IEEE 802.3u 100BASE-X
- IEEE 802.3x Flow control - full-duplex operation
- IEEE 802.3z 1000BASE-X

IPv4 Features

- RFC 768 User Datagram Protocol (UDP)
- RFC 791 Internet Protocol (IP)
- RFC 792 Internet Control Message Protocol (ICMP)
- RFC 793 Transmission Control Protocol (TCP)
- RFC 826 Address Resolution Protocol (ARP)
- RFC 894 Standard for the transmission of IP datagrams over Ethernet networks
- RFC 919 Broadcasting Internet datagrams
- RFC 922 Broadcasting Internet datagrams in the presence of subnets
- RFC 932 Subnetwork addressing scheme
- RFC 950 Internet standard subnetting procedure
- RFC 951 Bootstrap Protocol (BootP)
- RFC 1027 Proxy ARP
- RFC 1035 DNS client
- RFC 1042 Standard for the transmission of IP datagrams over IEEE 802 networks
- RFC 1071 Computing the Internet checksum
- RFC 1122 Internet host requirements
- RFC 1191 Path MTU discovery
- RFC 1256 ICMP router discovery messages

- RFC 1518 An architecture for IP address allocation with CIDR
- RFC 1519 Classless Inter-Domain Routing (CIDR)
- RFC 1542 Clarifications and extensions for BootP
- RFC 1591 Domain Name System (DNS)
- RFC 1812 Requirements for IPv4 routers
- RFC 1918 IP addressing
- RFC 2581 TCP congestion control

IPv6 Features

- RFC 1981 Path MTU discovery for IPv6
- IPv6 specification
- RFC 2464 Transmission of IPv6 packets over Ethernet networks
- RFC 3056 Connection of IPv6 domains via IPv4 clouds
- RFC 3484 Default address selection for IPv6
- RFC 3596 DNS extensions to support IPv6
- RFC 4007 IPv6 scoped address architecture
- RFC 4193 Unique local IPv6 unicast addresses
- RFC 4291 IPv6 addressing architecture
- RFC 4443 Internet Control Message Protocol (ICMPv6)
- RFC 4861 Neighbor discovery for IPv6
- RFC 4862 IPv6 Stateless Address Auto-Configuration (SLAAC)
- RFC 5014 IPv6 socket API for source address selection
- RFC 5095 Deprecation of type 0 routing headers in IPv6
- RFC 5175 IPv6 Router Advertisement (RA) flags option
- RFC 6105 IPv6 Router Advertisement (RA) guard

Management

- AT Enterprise MIB including AMF Plus MIB and SNMP traps
- SNMPv1, v2c and v3
- IEEE 802.1AB Link Layer Discovery Protocol (LLDP)
- RFC 1155 Structure and identification of management information for TCP/IP-based Internets
- RFC 1157 Simple Network Management Protocol (SNMP)
- RFC 1212 Concise MIB definitions
- RFC 1213 MIB for network management of TCP/IP-based Internets: MIB-II
- RFC 1215 Convention for defining traps for use with the SNMP
- RFC 1227 SNMP MUX protocol and MIB
- RFC 1239 Standard MIB
- RFC 1724 RIPv2 MIB extension
- RFC 2011 SNMPv2 MIB for IP using SMIv2
- RFC 2012 SNMPv2 MIB for TCP using SMIv2
- RFC 2013 SNMPv2 MIB for UDP using SMIv2
- RFC 2096 IP forwarding table MIB
- RFC 2578 Structure of Management Information v2 (SMIv2)
- RFC 2579 Textual conventions for SMIv2
- RFC 2580 Conformance statements for SMIv2
- RFC 2674 Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN extensions
- RFC 2741 Agent extensibility (AgentX) protocol
- RFC 2787 Definitions of managed objects for VRRP
- RFC 2819 RMON MIB (groups 1,2,3 and 9)
- RFC 2863 Interfaces group MIB
- RFC 3164 Syslog protocol
- RFC 3411 An architecture for describing SNMP management frameworks
- RFC 3412 Message processing and dispatching for the SNMP
- RFC 3413 SNMP applications
- RFC 3414 User-based Security Model (USM) for SNMPv3
- RFC 3415 View-based Access Control Model (VACM) for SNMP
- RFC 3416 Version 2 of the protocol operations for the SNMP
- RFC 3417 Transport mappings for the SNMP
- RFC 3418 MIB for SNMP
- RFC 3621 Power over Ethernet (PoE) MIB
- RFC 3635 Definitions of managed objects for the Ethernet-like interface types
- RFC 3636 IEEE 802.3 MAU MIB
- RFC 4188 Definitions of managed objects for bridges
- RFC 4318 Definitions of managed objects for bridges with RSTP
- RFC 4560 Definitions of managed objects for remote ping, traceroute and lookup operations
- RFC 6527 Definitions of managed objects for VRRPv3

RFC 7011 IPFIX: a method of exporting IP flow data in a network for analysis

Multicast Support

- IGMP query solicitation
- IGMP snooping (IGMPv1, v2 and v3)
- IGMP snooping fast-leave
- MLD snooping (MLDv1 and v2)
- RFC 2715 Interoperability rules for multicast routing protocols
- RFC 4541 IGMP and MLD snooping switches

Quality of Service (QoS)

- IEEE 802.1p Priority tagging
- IEEE 802.1Qaz Enhanced Transmission Selection (ETS)
- RFC 2211 Specification of the controlled-load network element service
- RFC 2474 DiffServ precedence for eight queues/port
- RFC 2475 DiffServ architecture
- RFC 2597 DiffServ Assured Forwarding (AF)
- RFC 2697 A single-rate three-color marker
- RFC 2698 A two-rate three-color marker
- RFC 3246 DiffServ Expedited Forwarding (EF)

Resiliency Features

- IEEE 802.1AX Link aggregation (static and LACP)
- IEEE 802.1D MAC bridges
- IEEE 802.1s Multiple Spanning Tree Protocol (MSTP)
- IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)
- IEEE 802.3ad Static and dynamic link aggregation
- RFC 5798 Virtual Router Redundancy Protocol version 3 (VRRPv3) for IPv4 and IPv6

Routing Information Protocol (RIP)

- RFC 1058 Routing Information Protocol (RIP)
- RFC 2081 RIPng protocol applicability statement
- RFC 2082 RIP-2 MD5 authentication
- RFC 2453 RIPv2

Security Features

- SSH remote login
- SSLv2 and SSLv3
- TACACS+ Accounting and Authentication
- IEEE 802.1X Authentication protocols (TLS, TTLS, PEAP and MD5)
- IEEE 802.1X Multi-supplicant authentication
- IEEE 802.1X Port-based network access control
- RFC 2818 HTTP over TLS ("HTTPS")
- RFC 3546 Transport Layer Security (TLS) extensions
- RFC 3579 RADIUS support for Extensible Authentication Protocol (EAP)
- RFC 3748 PPP Extensible Authentication Protocol (EAP)
- RFC 4251 Secure Shell (SSHv2) protocol architecture
- RFC 4252 Secure Shell (SSHv2) authentication protocol
- RFC 4253 Secure Shell (SSHv2) transport layer protocol
- RFC 4254 Secure Shell (SSHv2) connection protocol
- RFC 8446 Transport Layer Security (TLS) v1.3
- RFC 8894 Simple Certificate Enrollment Protocol (SCEP)

Services

- RFC 854 Telnet protocol specification
- RFC 855 Telnet option specifications
- RFC 857 Telnet echo option
- RFC 858 Telnet suppress go ahead option
- RFC 1091 Telnet terminal-type option

- RFC 1350 Trivial File Transfer Protocol (TFTP)
- RFC 1985 SMTP service extension
- RFC 2049 MIME
- RFC 2131 DHCPv4 (server, relay and client)
- RFC 2132 DHCP options and BootP vendor extensions
- RFC 2616 Hypertext Transfer Protocol - HTTP/1.1
- RFC 2821 Simple Mail Transfer Protocol (SMTP)
- RFC 2822 Internet message format
- RFC 3046 DHCP relay agent information option (DHCP option 82)
- RFC 3315 DHCPv6 (server, relay and client)
- RFC 3633 IPv6 prefix options for DHCPv6
- RFC 3646 DNS configuration options for DHCPv6
- RFC 3993 Subscriber-ID suboption for DHCP relay agent option
- RFC 4330 Simple Network Time Protocol (SNTP) version 4
- RFC 5905 Network Time Protocol (NTP) version 4

VLAN Support

- Generic VLAN Registration Protocol (GVRP)
- IEEE 802.1Q Virtual LAN (VLAN) bridges
- IEEE 802.1v VLAN classification by protocol and port

Voice over IP (VoIP)

- LLDP-MED ANSI/TIA-1057
- Voice VLAN

Feature Licenses

	Description	Includes	Stack Licensing
AT-FL-G98MX-CP	Continuous PoE license	■ Continuous PoE power for PSm and HSm models	■ One license per stack member
AT-FL-G98MX-UD	Mixed stacking license	■ UniDirectional Link Detection	■ One license per stack member

ORDERING INFORMATION

AT-GS980MX/10HSm-xx	8-port 100M/1/2.5/5G PoE++ stackable switch with 2 SFP+ ports and a single fixed power supply
AT-GS980MX/18HSm-xx	16-port 100M/1/2.5/5G PoE++ stackable switch with 2 SFP+ ports and a single fixed power supply
AT-GS980MX/28-xx	24-port 10/100/1000T stackable switch with 4 SFP+ ports and a single fixed power supply
AT-GS980MX/28PSm-xx	20-port 10/100/1000T PoE+ and 4-ports 100M/1/2.5/5G PoE+ stackable switch with 4 SFP+ ports and a single fixed power supply
AT-GS980MX/52-xx	48-port 10/100/1000T stackable switch with 4 SFP+ ports and a single fixed power supply
AT-GS980MX/52PSm-xx	40-port 10/100/1000T PoE+ and 8-ports 100M/1/2.5/5G PoE+ stackable switch with 4 SFP+ ports and a single fixed power supply
AT-RKMT-J15	Rack mount shelf kit for GS980MX/10HSm
AT-BRKT-J24	Wall mount kit for GS980MX/10HSm
AT-BRKT-J22	Wall-mount kit for GS980MX/28 & 52
AT-VT-Kit3	Management Cable (USB to Serial Console)

Where xx = 10 forUS power cord
20 for no power cord
30 for UK power cord
40 for AU power cord
50 for EU power cord

Accessories

10G SFP+ Modules	
AT-SP10TM	1G/2.5G/5G/10G, 100m copper, TAA ¹
AT-SP10SR	10GSR 850 nm short-haul, 300 m with MMF
AT-SP10SR/I	10GSR 850 nm short-haul, 300 m with MMF industrial temperature
AT-SP10LRa/I	10GBASE-LR, 1310 nm, 10 km with SMF, I-Temp, TAA ¹
AT-SP10BD10/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 10 km industrial temperature, TAA ¹
AT-SP10BD10/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 10 km industrial temperature, TAA ¹
AT-SP10BD20-12	10G Bi-Di, 1270 nm TX/1330 nm RX, 20km, TAA ¹
AT-SP10BD20-13	10G Bi-Di, 1330 nm TX/1270 nm RX, 20km, TAA ¹
AT-SP10BD40/I-12	10 GbE Bi-Di (1270 nm Tx, 1330 nm Rx) fiber up to 40 km industrial temperature, TAA ¹
AT-SP10BD40/I-13	10 GbE Bi-Di (1330 nm Tx, 1270 nm Rx) fiber up to 40 km industrial temperature, TAA ¹
10GbE SFP+ Cables	
AT-SP10TW1	1 meter SFP+ direct attach cable
AT-SP10TW3	3 meter SFP+ direct attach cable

1000Mbps SFP Modules	
AT-SPTX	1000T 100 m copper
AT-SPTX/I	1000SX GbE multi-mode 850 nm fiber up to 550 m industrial temperature
AT-SPSX	1000SX GbE multi-mode 850 nm fiber up to 550 m
AT-SPEX	1000X GbE multi-mode 1310 nm fiber up to 2 km
AT-SPSX/I	1000SX GbE multi-mode 850 nm fiber up to 550 m industrial temperature
AT-SPLX10a	1000LX SFP, LC, SMF, 1310nm (10km), TAA ¹
AT-SPLX10/I	1000LX GbE single-mode 1310 nm fiber up to 10 km industrial temperature
AT-SPBD10-13	1000LX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 10 km
AT-SPBD10-14	1000LX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 10 km
AT-SPBD40-13/I	1000LX GbE single-mode Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 40 km, industrial temperature
AT-SPBD40-14/I	1000LX GbE single-mode Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 40 km, industrial temperature
AT-SPLX40	1000LX GbE single-mode 1310 nm fiber up to 40 km

¹ Trade Act Agreement compliant (TAA)